

Technology Stack

Date	27 June 2026
Team ID	xxxxxxx
Project Name	EduGenie Google Gemini Powered Learning Assistant
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side (e.g., React, Angular, HTML/CSS)	Streamlit, HTML, CSS	Streamlit enables rapid development of an interactive learning interface with minimal code. Custom HTML and CSS improve the visual design and student experience.
2	Backend / Server-Side (e.g., Node.js, Python/Django, Java)	Python, FastAPI	Python supports AI integration, while FastAPI provides high-performance REST APIs, automatic Swagger documentation, and a clean connection between the interface and Gemini services.
3	Database / Data Storage (e.g., MySQL, MongoDB, PostgreSQL)	JSON Files (history.json, feedback.json)	JSON files provide lightweight local storage for learner history, saved interactions, quiz results, and feedback without a dedicated database server.
4	Cloud / Hosting / Deployment (e.g., AWS, Azure, Heroku, Docker)	Local Deployment using Uvicorn and Streamlit	The solution is deployed locally for development and demonstration. Its modular design can be migrated to a cloud platform in future versions.
5	Version Control & CI/CD (e.g., GitHub, GitLab, Jenkins)	Git, GitHub	Git and GitHub support source-code management, version tracking, collaboration, and secure project backup.
6	Third-Party APIs / Other Tools (e.g., Stripe, Twilio, Google Maps)	Google Gemini API, pytest, ReportLab	Google Gemini API generates learning content and feedback. PyTest supports automated testing, and ReportLab generates downloadable learning reports.